B. Tech.(Computer Engg.) V Semester Examinations

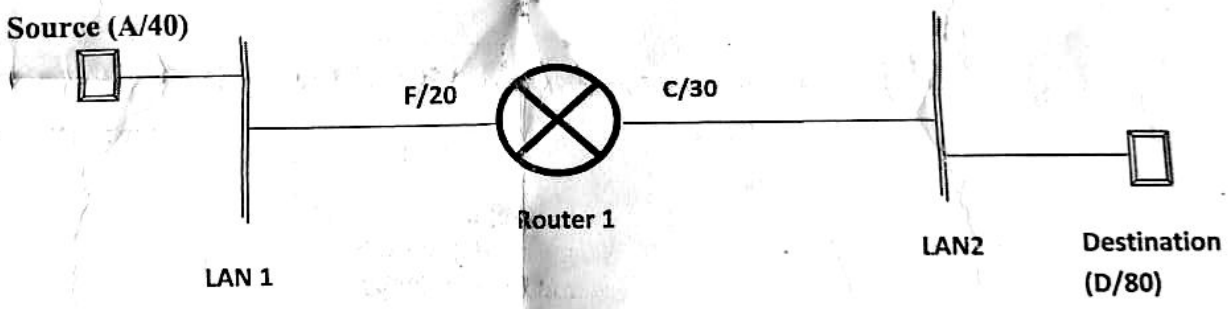
Session 2024-25

Time: 1 hr. **Computer Networks (CEN - 505) Sessional-I** Max. Marks: 15

Note: Attempt All Questions

1. CO1 The transport layer protocols used for real time multimedia, file transfer, DNS and email, respectively are 3
(A) TCP, UDP, UDP and TCP
(B) UDP, TCP, TCP and UDP
(C) UDP, TCP, UDP and TCP
(D) TCP, UDP, TCP and UDP
Justify your answer.

2. CO1 Suppose a user wants to send a data from a node with logical address 'A' and physical address 40, located on one local area network, to a node with a logical address 'D' and physical address 80, located on another local area network. If the data coming from the upper layers has service point address 'i' (Assume 'i' is the port address of the sending process and 'j' is the port address of the receiving process at receiver machine) and the size of the data is triple than the network layer can handle. Show the scenario, how addressing is done at each layer of OSI model from source to destination.



3. CO1 An ISP is granted a block of addresses starting with 190.100.0.0/16. The ISP needs to distribute these addresses to four groups of customers, A, B, C and D requesting 8000, 16000, 4000 and 2000 addresses respectively. Design the sub blocks and find out how many addresses are still available after these allocations. 4

4. CO2 What are the major problems in NRZ line encoding scheme and how can it be handled? Sketch the HDB3 and 2B1Q encoding for the following bit stream, assume that the number of 1's so far are odd and last signal was at low level: 4

1100000100000001011